Sushil Thakur

Full-Stack .NET Developer | ASP.NET Core | Angular | RAG & LLM Integration | C# & Python
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Profile Summary

Full-stack .NET Developer with 4 years of experience building enterprise applications with C#, ASP.NET Core REST APIs, Angular, and SQL databases, plus hands-on experience with AI-powered features. Most recently worked on a RAG-based support chatbot and role-based applications for Video Management System (VMS) products in the R&D team at CP Plus. Strong in SOLID principles, Clean Architecture, debugging, and performance optimization, with working knowledge of Docker, CI/CD, and Azure. Contributed to 27+ software projects across Smart City, Video Surveillance, IoT, and Enterprise domains.

Technical Skills

- **Languages:** C#, Python, SQL
- **Backend and Frameworks:** .NET 8/10+, ASP.NET Core, MVC, REST API, SOAP, CQRS, Minimal API
- **Frontend:** Angular, TypeScript, Bootstrap, HTML5, CSS3, JavaScript, Figma
- **AI/LLM Integration:** Azure OpenAI, Azure AI Services, Retrieval-Augmented Generation (RAG), LLM Integration, Model Context Protocol (MCP)
- **Databases & Messaging:** MSSQL, PostgreSQL, SQLite, MySQL, InfluxDB, RabbitMQ
- **Cloud & DevOps:** Microsoft Azure, Docker, Git, GitHub, GitLab, CI/CD, Jira, Postman
- **Architecture & Practices:** Clean Architecture, SOLID Principles, MVVM, CQRS, Repository Pattern, Unit Testing, Performance Optimization, Debugging & Logging, Multithreading, Agile
- **Desktop Development:** WPF, Avalonia UI, MVVM Architecture, Dependency Injection, XAML, Custom Controls

Professional Experience

Software Engineer – R&D (Contract) | CP Plus *Mar 2026 – Sep 2026*
Noida, India

Project 1: VMS (Video Management System)

- **AI Support Chatbot (RAG):** Created a RAG-based support chatbot within the VMS Admin app, retrieving from system documentation to answer user queries about configuration and troubleshooting.
- **Multi-App Desktop Architecture:** Built two role-based desktop applications (Admin + Operator) in Avalonia UI (.NET 10) using MVVM with CommunityToolkit source generators and DI-based service composition.
- **Live Camera Video Wall:** Crafted an infinite-scroll multi-camera video wall with drag-and-drop placement, resizable grid cells, full-screen mode, and live per-cell status overlays.
- **ONVIF Integration:** Implemented ONVIF protocol support for automated camera discovery and PTZ (pan-tilt-zoom) control across connected devices.
- **Role-Based Access & User Management:** Built full CRUD UI for users and roles with permission-key-based RBAC, gating access and auto-navigation by assigned role.
- **Tech Stack:** C#/.NET 10, Avalonia UI, MVVM, SignalR, ONVIF, RAG, JWT authentication

Project 2: CP DMT (Device Management Tool)

- **Multi-Step Production Workflow UI:** Established a structured desktop workflow (Scan → Flash → Burn → Functional Test → Boot Test → Print Label) that reduced operator errors by enforcing sequential step completion.
- **Firmware Flashing & Device Discovery:** Constructed a real-time network camera scanning panel and an interactive Flash panel with progress tracking, MD5 firmware verification, and cancelable async operations.
- **Serial & MAC Burn + Label Printing:** Added a Burn panel to write Serial Number/MAC Address to devices with label preview and direct printer integration.
- **Tech Stack:** C#/.NET, Avalonia UI, MVVM, async/await, JWT authentication

Software Engineer | Ajeevi Technologies Pvt Ltd *Dec 2023 – Feb 2026*
Noida, India

Project 1: ICCC-Smart City (Integrated Command & Control Center)

- **High-Scale IoT Data:** Engineered real-time GPS dashboards and backend modules processing live telemetry from 1,000+ IoT devices with sub-second reporting latency.
- **Modern API Architecture:** Designed high-performance backend services with Minimal APIs on .NET Core, reducing boilerplate and improving endpoint execution speed.

- **Multi-Database Orchestration:** Integrated MSSQL/MySQL for relational data with InfluxDB for high-velocity time-series sensor data.
- **Optimization:** Assembled a reusable backend component library, cutting development time for new features by approximately 20%.

Project 2: Video Management System (VMS)

- **Architectural Implementation:** Applied CQRS to separate read/write operations, improving API responsiveness and data consistency.
- **Performance Tuning:** Conducted deep-level MSSQL optimization, reducing query execution time for large video metadata sets.

Project 3: Smart Parking Management System (SPMS)

- **Scalable Service Design:** Produced RESTful APIs and modularized backend services enabling horizontal scaling as parking site density increases.
- **Real-Time Synchronization:** Optimized MySQL/InfluxDB queries to handle simultaneous occupancy updates from hundreds of concurrent sensors.
- **Admin Dashboard:** Built an Angular-based admin monitoring dashboard displaying real-time parking occupancy and sensor health status across all sites.

Software Engineer Intern | So Infotech Pvt Ltd *Jan 2023 – Dec 2023*
Noida, India

Project: Concert e-Ticket Booking (Enterprise Ticket Management System)

- **End-to-End Development:** Authored the core ticket-booking engine using ASP.NET Core MVC and SQL Server, focusing on transaction integrity during peak booking windows.
- **API Integration:** Delivered secure REST APIs and user authentication modules.

Education

Master of Computer Applications (MCA) — Mangalayatan University, Aligarh, UP | CGPA: 8.4

Certifications & Courses

Microsoft Certified: Azure Developer Associate — Microsoft | Issued 2026
Generative AI for .NET Developers with Azure AI Services — Udemy | Issued 2026